

Introduction to Manifolds

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Abstract

Contents

1	Introduction to Manifolds	3
1.1	Pseudogroup of Transformations	3
1.2	Manifolds and their Structures	3
1.3	Maps between Manifolds	5
2	New Manifolds from Old	6
2.1	Submanifolds	6
2.2	Product Manifolds	6
2.3	Quotient Manifolds	6
3	Real Manifolds	7
3.1	Topological Properties of Real Manifolds	7
3.2	Differentiable Structure on Real Manifolds	7
3.3	Analytic Structure on Real Manifolds	7
3.4	The Low Dimension Case	7
3.5	Manifolds with Boundaries	8

1 Introduction to Manifolds

1.1 Pseudogroup of Transformations

Definition 1.1.1 (The Set of Homeomorphisms of Open Subsets)

Let $(X, \mathcal{T})$ be a topological space. Define the set of homeomorphisms of X between open subsets to be

$$H(X) = \{f : U \rightarrow V \mid U, V \in \mathcal{T}, f \text{ is a homeomorphism}\}$$

We establish the definition of manifolds in a slightly non-standard way via the use of pseudogroups.

Definition 1.1.2 (Pseudogroup of Transformations)

Let $(X, \mathcal{T})$ be a topological space. A pseudogroup of transformations on X is a subset of homeomorphisms

$$\Gamma \subseteq H(X)$$

such that the following are true.

- (Cover) There exists an indexing set I and $f_i : U_i \rightarrow V_i \in \Gamma$ for $i \in I$ such that $X = \bigcup_{i \in I} U_i$.
- (Restriction) If $f : U \rightarrow V \in \Gamma$ and $W \subseteq U$, then $f|_W \in \Gamma$.
- (Local) If $f : U \rightarrow V \in H(X)$ and there exists an open cover $U = \bigcup_{i \in I} U_i$ such that $f|_{U_i} \in \Gamma$, then $f \in \Gamma$.
- (Inverse) If $f \in \Gamma$ then $f^{-1} \in \Gamma$.
- (Composition) If $f_1 : U_1 \rightarrow V_1 \in \Gamma$ and $f_2 : V_1 \rightarrow W_2 \in \Gamma$ such that $V_1 \cap V_2 \neq \emptyset$, then

$$f_2 \circ f_1 : f_1^{-1}(V_1 \cap V_2) \rightarrow f_2(V_1 \cap V_2)$$

lies in Γ

Manifolds is more or less some sort of a surface living in $\mathbb{R}^n$. To translate notions of differentiability to a manifold we want to cover the manifold with open sets isomorphic to $\mathbb{R}^m$. The pseudogroup of transformation is essentially determining what structure we want to investigate.

Denote Γ_{C^r} the pseudogroup where all transformations are C^r . Different values of r encode different information on the manifold since the underlying transformations between the open sets that cover the manifold change.

Definition 1.1.3 (C^r -Differentiable Maps) Let $U, V \subseteq \mathbb{R}^n$ be open. Denote $C^r(U, V)$ the set of all r times differentiable maps from U to V such that the r th derivative is continuous. By convention, $C^0(U, V)$ is the set of all continuous maps from U to V and $C^k(\mathbb{R}^n) = C^k(\mathbb{R}^n, \mathbb{R}^n)$. $C^\infty(U, V)$ denotes the set of all infinitely differentiable (smooth) maps from U to V .

Example 1.1.4

The following are pseudogroups of transformations.

- $\Gamma(\mathbb{R}^n, C^0) = H(\mathbb{R}^n)$.
- $\Gamma(\mathbb{R}^n, C^k) = \{f : U \rightarrow V \in H(\mathbb{R}^n) \mid f \in C^k(U, V) \text{ and } f^{-1} \in C^k(V, U)\}$ for $1 \leq k \leq \infty$.
- $\Gamma(\mathbb{R}^n, C^\infty) = \{f : U \rightarrow V \in H(\mathbb{R}^n) \mid f \in C^\infty(U, V) \text{ and } f^{-1} \in C^\infty(V, U)\}$.
- $\Gamma(\mathbb{R}^n, \mathcal{A}) = \{f : U \rightarrow V \in H(\mathbb{R}^n) \mid f, f^{-1} \text{ are analytic}\}$.
- $\Gamma(H^n, C^\infty) = \{f : U \rightarrow V \in H(H^n) \mid f \in C^\infty(U, V) \text{ and } f^{-1} \in C^\infty(V, U)\}$.

1.2 Manifolds and their Structures

Definition 1.2.1 (Atlas)

Let M be a topological space. An atlas of M compatible with a pseudogroup of transformations Γ is a family of pairs $\{(U_i, \phi_i) \mid i \in I\}$, called charts, such that the following are true.

- $\{U_i \mid i \in I\}$ is an open cover of M .
- Each ϕ_i is a homeomorphism of U_i onto an open set of S .
- Whenever $U_i \cap U_j$ is nonempty, the mapping $\phi_j \circ \phi_i^{-1}$ of $\phi_i(U_i \cap U_j)$ onto $\phi_j(U_i \cap U_j)$ is an element of Γ .

This definition complements well with our explanation. For an atlas to make sense with a pseudogroup, the target (range) should be the underlying $\mathbb{R}^m$. The U_i should at least cover M and be part of M . And finally areas with overlapping should have maps that allow transition between two covers, which is what the third condition is saying.

Definition 1.2.2 (Compatible Charts) Let (U, ϕ) be a chart of a space M . We say that it is compatible with another chart (V, φ) if $\varphi \circ \phi^{-1} : \phi(U \cap V) \rightarrow \varphi(U \cap V)$ is in Γ . We say that (U, ϕ) is compatible with an atlas $\{(U_i, \phi_i) \mid i \in I\}$ of M if (U, ϕ) is compatible with every chart in the atlas.

To prove that two atlases A and A' are not equal, one should find a chart in A that is incompatible with a chart in A' , or vice versa.

Definition 1.2.3 (Complete Atlas) A complete atlas on a topological space M compatible with Γ is an atlas of M compatible with Γ which is not contained in any other atlas M compatible with Γ .

Complete atlases are also called maximal atlases.

Lemma 1.2.4 Every atlas of M compatible with Γ is contained in a unique complete atlas.

Proof Let $A = \{(U_i, \phi_i)\}$ be an atlas of M compatible with Γ . Let $\tilde{A}$ be the family of all pairs (U_i, ϕ_i) such that $\phi : U_i \subseteq M \rightarrow S$ is a homeomorphism such that $\phi_j \circ \phi_i^{-1} : \phi_i(U_i \cap U_j) \rightarrow \phi_j(U_i \cap U_j)$ is an element of Γ whenever $U_i \cap U_j$ is nonempty. Then $\tilde{A}$ is the complete atlas of M containing A . ■

However note that this is not to say that M has a unique Γ -structure.

By collecting atlas and using the complete atlas of a topological space, we can talk about all possible transitions between the open covers.

Definition 1.2.5 (Manifolds)

Let X be a space. Let Γ be a pseudogroup of transformations over X . A Γ -manifold is a pair $(M, \mathcal{A})$ where

- M is a second countable Hausdorff space.
- $\mathcal{A}$ is a complete atlas compatible with Γ .

A chart on a real manifold M of dimension n is actually an open set together with n coordinate functions. Indeed if (U, ϕ) is a chart of M , then $\phi : U \rightarrow \mathbb{R}^n$ is a map to $\mathbb{R}^n$. If we are interested in the particular components of ϕ , we write $\phi = (x^1, \dots, x^n)$ where $x^i : U \rightarrow \mathbb{R}$ is the coordinate function sending a point on U to its i th coordinate. All the x^i 's combine to recover ϕ . If we would like to talk about the actual coordinate on $\mathbb{R}^n$ instead of the coordinate function, we write x_i instead.

Definition 1.2.6 (Transition Functions)

Let M be a manifold. Let $\{(U_i, \phi_i) \mid i \in I\}$ be an atlas on M . The transition functions on M are the functions

$$\phi_j \circ \phi_i^{-1} : \phi_i(U_i \cap U_j) \rightarrow \phi_j(U_i \cap U_j)$$

for $U_i \cap U_j \neq \emptyset$.

We now discuss some topological properties of manifolds whose charts are locally $\mathbb{R}^n$, $\mathbb{C}^n$ or H^n .

1.3 Maps between Manifolds

Definition 1.3.1 (Morphism of Manifolds)

Let Γ be a pseudogroup of transformations. Let M, N be Γ -manifolds whose atlases are given by $\{(U_i, \phi_i) \mid i \in I\}$ and $\{(V_j, \psi_j) \mid j \in J\}$ respectively. Let $f : M \rightarrow N$ be a function. We say that f is a morphism if

$$\psi_j \circ f \circ \phi_i^{-1} \in \Gamma$$

whenever $f(U_i) \cap V_j \neq \emptyset$.

Lemma 1.3.2

Let M, N, K be manifolds. Let $f : M \rightarrow N$ and $g : N \rightarrow K$ be morphisms. Then $g \circ f$ is a morphism.

Lemma 1.3.3

Let M be a manifold. Then $\text{id}_M : M \rightarrow M$ is a morphism.

Definition 1.3.4 (Isomorphisms)

Let M, N be manifolds. Let $f : M \rightarrow N$ be a morphism. We say that f is an isomorphism if there exists a morphism $g : N \rightarrow M$ such that

$$g \circ f = \text{id}_M \quad \text{and} \quad f \circ g = \text{id}_N$$

Definition 1.3.5 (The Group of Automorphisms)

Let M be a manifold. Define the group of automorphisms of M to be the set

$$\text{Aut}(M) = \{f : M \rightarrow M \mid f \text{ is an isomorphism} \}$$

2 New Manifolds from Old

2.1 Submanifolds

Definition 2.1.1 (Embedded Submanifolds)

Let M be a manifold. An embedded submanifold of M is a subset $N \subseteq M$ such that for all $p \in N$, there exists a chart $(U, \phi = (x^1, \dots, x^n))$ of M at p such that $x_{i_1}|_{U \cap N} = \dots = x_{i_{n-k}}|_{U \cap N} = 0$ vanishes as functions for some fixed $k \in \mathbb{N}$.

Intuitively, the condition that some coordinates function of charts on N vanish means that N really have charts in $\mathbb{R}^{n-m}$ instead of $\mathbb{R}^n$. This condition is more commonly written as

$$\phi(U \cap N) = \phi(U) \cap \mathbb{R}^k \times \{0\}^{n-k}$$

Proposition 2.1.2

Let M be a manifold. Let $N \subseteq M$ be an embedded submanifold. Then the following are true.

- N is a manifold in its own right whose atlas is given by

$$\{(U_i \cap N, \phi|_{U_i \cap N}) \mid i \in I\}$$

where $\{(U_i, \phi_i) \mid i \in I\}$ is the atlas of M .

- If $n - k$ coordinate function of M vanishes at N , then $\dim(N) = k$.
- The inclusion map $N \hookrightarrow M$ is a morphism of manifolds.

It is not possible to characterize embedded submanifolds using simply inclusions $N \hookrightarrow M$.

Example 2.1.3 (Alexander Horned Ball)

The Alexander horned ball is a topological 3-manifold equipped with an inclusion map of sets to S^3 , but is not an embedded submanifold of S^3 .

2.2 Product Manifolds

Proposition 2.2.1

Let M, N be manifolds. Then the following are true.

- $M \times N$ is a manifold of dimension.
- If (U, ϕ) is a chart of M and (V, ψ) is a chart of N , then $(U \times V, \phi \times \psi)$ is a chart of $M \times N$.

2.3 Quotient Manifolds

Proposition 2.3.1

Let G be a discrete group. Let M be a Γ -manifold that is a G -space. Suppose that the following are true.

- The action of G on M is free and proper.
- $g \cdot - : M \rightarrow M \in \text{Aut}(M)$ for all $g \in G$.

Then the following are true.

- The orbit space M/G is a Γ -manifold.
- $\dim(M/G) = \dim(M)$.

3 Real Manifolds

3.1 Topological Properties of Real Manifolds

Definition 3.1.1 (Types of Real Manifolds)

Let Γ be a pseudogroup of transformations over $\mathbb{R}^n$. Let M be a Γ -manifold.

- We say that M is a topological manifold if $\Gamma = \Gamma(\mathbb{R}^n, \mathcal{C}^0)$.
- Let $1 \leq k < \infty$. We say that M is a C^k -differentiable manifold if $\Gamma = \Gamma(\mathbb{R}^n, \mathcal{C}^k)$.
- We say that M is a smooth manifold if $\Gamma = \Gamma(\mathbb{R}^n, \mathcal{C}^\infty)$.
- We say that M is a real analytic manifold if $\Gamma = \Gamma(\mathbb{R}^n, \mathcal{A})$.

Definition 3.1.2 (Dimension)

Let Γ be a pseudogroup of transformations over $\mathbb{R}^n$. Let M be a Γ -manifold. Define the dimension of M to be $\dim(M) = n$.

Lemma 3.1.3

Let M be a manifold. Then M is locally compact.

Definition 3.1.4 (The Space of Differentiable Functions)

Let $k \in \mathbb{N} \cup \{\infty\}$. Let M be a C^k -differentiable manifold. Define the space of C^k -differentiable functions of M to be the set

$$\mathcal{C}^k(M) = \{f : M \rightarrow \mathbb{R} \mid f \text{ is a } C^k\text{-differentiable function}\}$$

together with pointwise addition and pointwise multiplication.

3.2 Differentiable Structure on Real Manifolds

Lemma 3.2.1 The following are true regarding the different types of manifolds.

- Every smooth manifold is a differentiable manifold of class C^r for any $r \in \mathbb{N}$
- Every differentiable manifold of class C^r is a differentiable manifold of class C^{r-1} for $r \geq 1$
- Every differentiable manifold of class C^r is a topological manifold

Proposition 3.2.2

Let $k \geq 1$. Let M be a C^k manifold with atlas $\mathcal{A}$. Then $\mathcal{A}$ is compatible with a unique C^∞ atlas.

ref: hirsch differential topology thm2.9

Proposition 3.2.3

Let $k \geq 1$. Let M, N be smooth manifolds. Then M and N are diffeomorphic if and only if M and N are C^k -diffeomorphic.

3.3 Analytic Structure on Real Manifolds

Theorem 3.3.1 (Grauert-Whitney)

Every smooth manifold admits a compatible real analytic structure.

3.4 The Low Dimension Case

Theorem 3.4.1 (Kervaire-Milnor)

Let M be a topological manifold of dimension $n \leq 3$. Then M admits a unique smooth structure.

Proposition 3.4.2

Let M, N be topological manifolds of dimension both 2 or 3. Then M and N are $\mathcal{C}^0$ -diffeomorphic if and only if M and N are diffeomorphic.

3.5 Manifolds with Boundaries**Definition 3.5.1** (Manifolds with Boundaries)

Let $k \in \mathbb{N} \cup \{\infty\}$. A $\mathcal{C}^k$ -manifold with boundary is a $\Gamma(H^n, \mathcal{C}^k)$ -manifold.

Lemma 3.5.2

Let $k \in \mathbb{N} \cup \{\infty\}$. Let M be a $\mathcal{C}^k$ -manifold. Then M is a $\mathcal{C}^k$ -manifold with boundary.

Definition 3.5.3 (Dimension)

Let M be a $\Gamma(H^n, \mathcal{C}^k)$ -manifold. Define the dimension of M to be $\dim(M) = n$.

Definition 3.5.4 (Boundary Charts)

Let M be a $\mathcal{C}^k$ -manifold with boundary. Let $p \in \partial M$. Let (U, ϕ) be a chart at p . We say that (U, ϕ) is a boundary chart at p if the following are true.

- $\phi(U) \subseteq \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_n \geq 0\}$.
- $\phi^{-1}(\{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_n = 0\}) \subseteq \partial M$.
- $\phi(p) = 0$.

Definition 3.5.5 (The Boundary of a Manifold)

Let M be a manifold with boundary. Define the boundary of M to be

$$\partial M = \{p \in M \mid p \text{ admits a boundary chart}\}$$

Lemma 3.5.6

Let M be a manifold with boundary. Then the following are true.

- $\partial M \cap M^\circ = \emptyset$.
- ∂M is a closed subset of M .

Proposition 3.5.7

Let M be a $\mathcal{C}^k$ -manifold with boundary. Then ∂M is a $\mathcal{C}^k$ -manifold of dimension $\dim(M) - 1$ with atlas

$$\{(U_\alpha|_{\partial M}, \phi_\alpha|_{\partial M})\}$$

where $\{(U_\alpha, \phi_\alpha)\}$ the atlas on M .